

CarcBlog Production Release, Feature & Performance QA Checklist

A detailed, production-ready checklist specifying **what to test** and **how to test** every feature, user workflow, performance SLA, search capability, and system resilience pattern before launching CarcBlog.

1. Core Feature & Functionality Testing

1.1 Editorial Platform & Article CMS

- ☐ **Rich Text Editing (BlockNote Editor):**
 - **What to test:** Headlines, paragraph formatting, bold, italic, strikethrough, inline code, H2/H3 headers, blockquotes, bullet lists, numbered lists, and code blocks with Shiki syntax highlighting.
 - **How to test:** Navigate to `/dashboard/articles/new`. Enter text for each formatting style. Verify live rendering in the editor canvas and saving to Sanity CMS.
 - ☐ **Media Uploads & Cloudinary CDN Integration:**
 - **What to test:** Cover image upload, inline image embedding, alt text assignment, and CDN resolution.
 - **How to test:** Upload a cover image in the article editor. Inspect network payload to ensure the URL points to Cloudinary/Sanity CDN (`https://cdn.sanity.io/...` or `https://res.cloudinary.com/...`). Verify images render with `loading="lazy"` and `decoding="async"`.
 - ☐ **Draft Auto-Saving & Scheduling:**
 - **What to test:** Article status lifecycle: `Draft` → `Scheduled` → `Published` → `Archived`.
 - **How to test:** Create an article. Click **Save Draft**. Verify status indicator displays `Draft`. Set a future publication date and click **Schedule**. Confirm entry displays under `/dashboard/articles?tab=scheduled`.
 - ☐ **Writer Dashboard Overview & Direct Article Links:**
 - **What to test:** Metrics overview (total views, readers, bookmarks, articles) and Recent Activity row links.
 - **How to test:** Navigate to `/dashboard`. Click on any article row under **Recent Activity**. Confirm it navigates directly to the live published article page (`/article/[slug]`) rather than opening CMS edit state.
 - ☐ **AI Writing Assistant Panel:**
 - **What to test:** AI prompt generation, headline/subtitle suggestions, article body expansion, and API key missing alert.
 - **How to test:** Open AI Assistant drawer in the article editor. Click **Quick Prompt Chips** (e.g. "Generate 5 viral headlines"). Click **Apply Headline & Subtitle** to auto-populate form fields. Click **Insert Full Story into Editor** to insert generated content into the BlockNote editor canvas.
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1.2 User Roles & RBAC Matrix Verification

Role	Permitted Actions	Gated / Restricted Actions	Test Procedure
Guest	Read public articles, view startup/founder/investor profiles, search global database, subscribe to newsletter	Cannot bookmark, like, comment, follow, or access /dashboard	Click bookmark/like button while unauthenticated → Verify redirect to /auth/sign-in or Clerk auth modal.
Registered User	+ Bookmark articles, post comments/replies, like articles, follow startups/writers, edit own profile (/u/[username])	Cannot create/edit articles or access CMS dashboard	Log in as reader. Verify /dashboard/articles returns 403 or redirects to /dashboard/profile .
Author / Writer	+ Access Writer Dashboard (/dashboard), create/edit own articles, upload media, manage drafts, view analytics	Cannot edit articles created by other writers or perform admin actions	Log in as writer. Create an article, publish it, and verify analytics update under /dashboard/analytics .

1.3 Interactive Social Features

- ☐ **Bookmarking:**
 - **What to test:** Persistence of bookmarked articles across client sessions.
 - **How to test:** Click **Bookmark** button on an article card or detail page. Refresh page — confirm bookmark icon remains highlighted. Navigate to </dashboard/bookmarks> — verify article appears in list. Click **Remove** — confirm item disappears.
- ☐ **Liking Articles:**
 - **What to test:** Optimistic UI update of article like counter.
 - **How to test:** Click **Like** button on an article. Confirm count increments immediately. Refresh page to verify persistent database increment in Supabase.
- ☐ **Nested Comment Threads:**
 - **What to test:** Top-level comment submission, nested replies, author badge, and timestamp formatting.
 - **How to test:** Submit a top-level comment. Submit a reply under another user's comment. Verify nested indentation, user avatar, and relative timestamp ([2 minutes ago](#)).
- ☐ **Social Sharing Modal:**
 - **What to test:** Native Web Share API fallback, Twitter/X share link, LinkedIn share link, and Copy Link to clipboard.
 - **How to test:** Click **Share** on an article. Test **Copy Link** — verify toast notification *"Link copied to clipboard!"*. Click Twitter/X button — verify correct URL and title parameters.

1.4 Structured Ecosystem Data Directories

- ☐ **Startups Directory (</startups> & [/startups/\[slug\]](/startups/[slug])):**

- **What to test:** 5,900+ startup database records, search filter, industry category normalization dropdown, region filter, pagination, and detail pages.
 - **How to test:**
 1. Visit </startups>. Verify grid displays 24 items with logos, names, normalized industry tags, HQ locations, and website links.
 2. Filter by industry dropdown: Select [AI \(1,243\)](#). Confirm grid filters instantly and URL updates to [?industry=AI](/?industry=AI).
 3. Filter by region dropdown: Select [Europe](#). Confirm grid filters to European startups ([?region=Europe](/?region=Europe)).
 4. Click a startup card (e.g. </startups/devfast>). Verify detailed company overview, HQ city/country, website link, and list of associated co-founders with job titles.
 - ☐ **Founders Directory (</founders> & [/founders/\[slug\]](/founders/[slug])):**
 - **What to test:** 11,000+ founder records, search filter, region filter, and detail pages.
 - **How to test:**
 1. Visit </founders>. Filter by search query (e.g., [Aadit Palicha](#)).
 2. Click founder profile (</founders/aadit-palicha>). Verify founder name, avatar, bio, location, LinkedIn link, and co-founded startup cards.
 - ☐ **Investor Directory (</investors> & [/investors/\[slug\]](/investors/[slug])):**
 - **What to test:** 5,000+ VC and angel investor records, investor type filter, target geography filter, and detail pages.
 - **How to test:**
 1. Visit </investors>. Filter by Investor Type ([VC](#), [Angel](#), [Accelerator](#)).
 2. Select Region [North America](#). Confirm results filter to North American investors.
 3. Click investor profile (</investors/406-ventures>). Verify firm thesis, first check size ([\\$500K - \\$2M](#)), investment stage, value add notes, team members, and application link.
 - ☐ **Funding Tracker (</funding>):**
 - **What to test:** Recent funding deals matrix, funding round badges ([Seed](#), [Series A](#), [Series B](#)), lead investor tags, and total capital counter.
 - **How to test:** Visit </funding>. Verify timeline orders by deal date descending. Test filtering by round stage.
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2. Global Search Infrastructure Testing

- ☐ **Command Palette ([Cmd+K](#) / [Ctrl+K](#)):**
 - **What to test:** Keyboard shortcuts, search modal trigger, REST API responses (</api/search>), cross-entity result grouping, and keyboard focus traps.
 - **How to test:**
 1. Press [Cmd+K](#) (Mac) or [Ctrl+K](#) (Windows) anywhere on the site. Confirm search modal opens instantly.
 2. Type [Stripe](#). Confirm results display separate sections for **Startups**, **Founders**, **Investors**, and **Articles**.
 3. Use [Down](#) and [Up](#) arrow keys to highlight items, and press [Enter](#) to navigate.
 4. Press [Esc](#) — confirm search modal closes cleanly.
- ☐ **URL Param Synchronization:**
 - **What to test:** Search query and filter persistence in browser history.

- **How to test:** Perform a search on `/startups?q=AI®ion=Europe`. Copy URL into a new tab — verify filters populate automatically from URL search params.

3. Performance & Core Web Vitals Verification

3.1 Mobile PageSpeed Insights SLA (Target Score ≥ 90)

Execute Google PageSpeed Insights on `https://carcblog-kappa.vercel.app/` using Emulated Moto G Power (Slow 4G):

- ☐ **Mobile Performance Score:** ≥ **90** (Target: 94).
- ☐ **First Contentful Paint (FCP):** < **1.8s** (Target: 1.2s).
- ☐ **Largest Contentful Paint (LCP):** < **2.5s** (Target: 1.8s).
- ☐ **Speed Index:** < **3.5s** (Target: 2.4s).
- ☐ **Cumulative Layout Shift (CLS):** < **0.1** (Target: 0.000).
- ☐ **Total Blocking Time (TBT):** < **100ms** (Target: 0ms).
- ☐ **Document TTFB / Latency:** < **50ms** via Vercel Edge Cache-Control headers (`public, s-maxage=3600`).

3.2 Database & Server Query Performance SLAs

Run `npx tsx scripts/profile-performance.ts` to verify database query execution times:

Query / Function	Target SLA	Verification Method
<code>getStartupIndustries()</code>	0 ms (Cached TTL)	Test 2nd consecutive invocation (returns cached category counts instantly)
<code>getInvestorTypes()</code>	0 ms (Cached TTL)	Test 2nd consecutive invocation
<code>/startups</code> (<code>getStartups</code>)	< 300 ms	Benchmark query with limit 24 + offset
<code>/startups/[slug]</code> (<code>getStartupBySlug</code>)	< 260 ms	Benchmark single-roundtrip nested join query
<code>/founders/[slug]</code> (<code>getFounderBySlug</code>)	< 250 ms	Benchmark single-roundtrip nested join query
<code>/investors/[slug]</code> (<code>getInvestorBySlug</code>)	< 250 ms	Benchmark single-roundtrip query
Search REST API (<code>/api/search</code>)	< 150 ms	Measure response time for <code>GET /api/search?q=AI</code>

4. Edge Cases, Error Handling & System Resilience

- ☐ **Missing Image Fallback Handling:**
 - **What to test:** Broken or null image URLs (`coverImage`, `logo_url`, `avatar_url`).

- **How to test:** Load an article or startup profile with a null/broken image. Verify image falls back cleanly to atmospheric gradient orb ([.placeholder-orb](#)) or initial letter avatar without broken image icons.
 - ☐ **Missing Database Records & 404 Routing:**
 - **What to test:** Non-existent slugs ([/startups/non-existent-slug](#), [/founders/invalid-id](#)).
 - **How to test:** Navigate to an invalid slug. Verify page handles error gracefully, either displaying custom 404 layout or redirecting cleanly back to directory index.
 - ☐ **Uncached / Cold-Start Resilience:**
 - **What to test:** First request after Vercel lambda cold start or cache invalidation.
 - **How to test:** In-memory cache handles cold starts gracefully, populating cache within 500ms and serving subsequent requests in 0ms.
 - ☐ **AI Assistant Key Missing Alert:**
 - **What to test:** Behavior when user has not configured AI API keys in settings.
 - **How to test:** Open AI Assistant panel without an API key. Confirm alert banner displays "Add an API key in Settings to enable AI generation" with link to [/dashboard/settings](#).
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5. SEO & Accessibility Standards (100 / 100 Standards)

5.1 SEO Verification

- ☐ **Dynamic Meta Tags:** Inspect DOM `<head>` on homepage, articles, startups, founders, and investors for unique `<title>` and `<meta name="description">`.
- ☐ **OpenGraph & Twitter Card Meta Tags:**
 - `og:title`, `og:description`, `og:image`, `og:url`, `og:type` (`article` or `website`).
 - `twitter:card` (`summary_large_image`), `twitter:title`, `twitter:image`.
- ☐ **Schema.org Structured Data:**
 - Test live article URL using Google Rich Results Test. Confirm `Article` JSON-LD schema renders author, publisher, and headline.
- ☐ **Dynamic Sitemap ([/sitemap.xml](#)) & Robots ([/robots.txt](#)):**
 - Visit [/sitemap.xml](#) — verify XML sitemap contains entries for all published articles, startup profiles, founder profiles, and investor profiles.
 - Visit [/robots.txt](#) — confirm `User-agent: *`, `Allow: /`, and `Sitemap: https://carcblog-kappa.vercel.app/sitemap.xml`.
- ☐ **Descriptive Link Text (SEO Audit 100):**
 - Verify zero generic link text ("View all", "Learn more", "click here", "href="#"). **All links must contain descriptive text or explicit** `aria-label`` attributes.

5.2 Accessibility & WCAG AA Compliance

- ☐ **Color Contrast Ratios:**
 - Footer copyright and legal links (`#cbd5e1` on `#0f172a` background): **8.5:1** contrast ratio (WCAG AA pass).
 - Primary text (`#0f172a` on `#ffffff`): **14:1** contrast ratio.
- ☐ **Strict Heading Hierarchy:**
 - Inspect document outline using Accessibility DevTools. Confirm strict H1 → H2 → H3 progression without skipped heading levels (e.g. no jumping H2 → H5).
- ☐ **Keyboard Nav & Focus Indicators:**

- Press **Tab** to navigate through navbar, buttons, dropdowns, and cards. Confirm visible focus ring outlines every focused element.
 - ☐ **ARIA Attributes:**
 - Confirm all icon-only buttons (Search trigger, mobile menu toggle, close buttons) have explicit `aria-label` attributes.
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6. Pre-Flight Release Checklist

1. ☐ **Environment Variables Audit (`.env`):**
 - `PUBLIC_SANITY_PROJECT_ID`
 - `PUBLIC_SANITY_DATASET`
 - `SANITY_API_TOKEN`
 - `PUBLIC_CLERK_PUBLISHABLE_KEY`
 - `CLERK_SECRET_KEY`
 - `DATABASE_URL` (Supabase Postgres database connection string)
2. ☐ **Database Migration Execution:**
 - Apply `db/migrations/001_create_startups_and_founders.sql`
 - Apply `db/migrations/002_create_investors.sql`
 - Apply `db/migrations/003_add_directory_indexes.sql` (b-tree performance indexes)
3. ☐ **Clean Production Build Verification:**
 - Execute `npm run build` locally — confirm zero Astro SSR build warnings or TypeScript compilation errors.
4. ☐ **Static Asset Caching Directives (`vercel.json`):**
 - Confirm `vercel.json` enforces `Cache-Control: public, max-age=31536000, immutable` headers for `/_astro/*` and static font files.